

JEE MAIN REVISION MASTER: ELLIPSE 2026 0 QUESTIONS 0% WEIGHTAGE ↓100%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 4 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Ellipse - Core Application	4 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Ellipse - Core Application (Moderate Yield — 4 PYQs)

Core Principle: Key problem pattern in Kcet Ellipse.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "The length of the latus rectum of $x^2+3y^2=12$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If the area of the ellipse is $\frac{\pi}{25} + \frac{\pi}{2}\lambda^2 = 1$ is 20π sq units, then λ is..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "The eccentricity of the ellipse $9x^2+25y^2=225$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

The length of the latus rectum of $x^2+3y^2=12$ is

- (A) $(2/3)$ units
- (B) $(1/3)$ units
- (C) $\frac{4\sqrt{3}}{3}$ units
- (D) 24 units

Correct Answer: None

Solution Analysis: $\frac{x^2}{12} + \frac{y^2}{4} = 1$ L.R = $\frac{2b^2}{a} = \frac{2 \times 4}{3} = \frac{8}{3} = \frac{4\sqrt{3}}{3}$